

This Month's DVD Contents

This month's DVD brings you GPU machine learning library and simulators along with an automation and measurement software for optical systems. If you are an enthusiast, there is a robotic research tool-kit for you. The top ten software in the DVD are mentioned below

PANKAJ V.

1. GPUMLib

GPUMLib is a C++ and CUDA-based high-performance GPU machine-learning library. It includes various machine-learning algorithms and components like back propagation (BP) and multi-back propagation (MBP), matrix factorisation and radial-basis function networks. It also provides machine-learning researchers and practitioners with an autonomous training system for creating BP and MBP networks. **Supported OS:** Windows. **Size:** 227MB

2. itom

itom is a lab-automation and measurement software designed for developing and running optical systems, but it is not limited to this field of application. itom is a versatile tool which can be used for building the control software for any measurement setup and features easy integration of different hardware like cameras or actuators through various plugins. itom is an open source free tool but you will find its functionality comparable to the commercial software packages like MATLAB or LabView. **Supported OS:** Windows. **Size:** Windows 32-bit: 118MB, Windows 64-bit: 168MB

3. Magic VLSI

Magic is an easy-to-use VLSI layout tool which has been popular with universities and small companies. It is an open source EDA alternative to commercial tools which will provide you with various useful features for your product design flows. The all new interactive maze router will allow you to specify hints to control routing. Also, with the multiple-windowing feature, you can view several portions of a circuit at the same time, or even portions of different circuits. **Supported OS:** Linux. **Size:** 3.6MB

4. Liquid PCB

Liquid PCB is an open source CAD application for designing PCBs. This simple tool has a unique interface with all the functionality and features available in just a few clicks. Also, it has a large, powerful file dialogue where all the recently visited directories are automatically remembered. Additionally, the one-menu system allows you to have all the functions in one menu. **Supported OS:** Windows. **Size:** 1.3MB

5. Mobile Robot Programming Toolkit

The Mobile Robot Programming Toolkit is an open source and extensive tool which can be used for performing robotics research to design and implement algorithms about simultaneous localisation and mapping, navigation and computer vision. It includes portable and well-tested applications with libraries covering data structures and algorithms employed in common robotics research areas. **Supported OS:** Windows, Linux. **Size:** Windows 32-bit: 102MB, Windows 64-bit: 112MB, Linux: 20.9MB

6. SVEditor

SVEditor, an Eclipse-based IDE written in Java, can be used for Verilog and SystemVerilog development. The interface is a colourising editor for SystemVerilog with support for content assist, source navigation, SystemVerilog source templates, source indent and auto-indent, and context-sensitive viewing of source documentation. **Supported OS:** Windows, Linux. **Size:** Windows 32-bit: 78.3MB, Windows 64-bit: 78.4MB, Linux: 77.3MB

7. QsapecNG

QsapecNG is an easy-to-use, Qt-based program for symbolic analysis of linear

analogue circuits. Its two independent parts, SapecNG and QsapecNG, are the framework and the gui of the application, respectively. This tool allows you to draw, solve and analyse analogue circuits and manage them as well. **Supported OS:** Windows, Linux, MacOSX. **Size:** Windows: 4.6MB, MacOSX: 15.9MB, Linux: 1.6MB

8. Gerber2PDF

Gerber2PDF is a simple command-line tool which converts Gerber files to PDF. It can convert multiple Gerber files at once and allows you to place the resulting layers on different pages within the PDF. Also, you can combine different layers onto a single page and render them with custom colours and transparency. **Supported OS:** Windows. **Size:** 311kB

9. gpsim

gpsim, a software simulator for PIC microcontrollers from Microchip, is fast with a simulation speed of about 20MHz. It allows real-time simulation to test PIC the same way as in a tangible universe. Gpsim's accurate design makes it comparable to a real PIC from the core to the I/O pins, including all their internal peripherals. **Supported OS:** Linux. **Size:** 2.4MB

10. CPUSim

CPU Sim is a Java application that allows users to design simple computer CPUs at microcode level and to run machine-language or Assembly-language programs on those CPUs through simulation. It is a useful tool for instructors who want their students to get hands-on exposure to a variety of architectures and a chance to design and implement their own architectures. **Supported OS:** Windows, Linux, MacOSX. **Size:** 7.3MB

The author is a technical journalist at EFY